



RECOVERY DECISION PLAN

## VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

# NSX Recovery Options & Clean-Rebuild Decision

Recovering the NSX overlay after the vCenter-redeploy VDS-UUID damage — what is verified, what is prescribed, and an eyes-open rebuild option.

NSX

VDS UUID

RECOVERY

REBUILD

DECISION

**v1.1**

VMware Cloud Foundation 9

# NSX Recovery Options & Clean-Rebuild Decision Plan

**Prepared by:** Virtual Control LLC **Date:** June 9, 2026 **Document Version:** 1.1 **Classification:** Internal

**Read this first — the honest headline.** There is **no Broadcom-prescribed "rebuild NSX config from scratch" procedure for VCF 9** that fits this situation. The VCF 9 NSX recovery docs cover only **restore from backup** (which re-imports the broken state) and **redploy of a failed Manager node** (no node has failed — the Manager is healthy). The prescribed route to fix the actual problem (a stale, un-clearable VDS-ownership record) is **Broadcom Support** (Corfu DB cleanup). This plan documents that, the options, and — if chosen eyes-open — the *phases* of a manual rebuild, which is **outside VCF orchestration** and is **not** a guess-free runbook.

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## 1. Verified Diagnosis (Recap)

- All 4 host transport nodes fail: NSX references VDS UUID `50 37 13 3f...` which no longer exists; the current VDS `vcenter-cl01-vds01` is `50 12 35 19...` (vCenter redeploy + VDS re-import gave a new UUID).
- Compute Manager: `REGISTERED_WITH_ERRORS` — error **90325 / 8012**, "Unable to clear NSX ownership" of the deleted VDS.

- Transport-node realization error **9560**: TZ "cannot be relocated from HostSwitch (old) to (new) in a single transaction."
- **Overlay data plane is down** — 0 tunnels — but NSX **management + control clusters are STABLE** and **mgmt/VLAN networking is healthy**.
- Config is **empty: 0 segments, 0 ports, 0 DFW rules, 0 edges** → almost nothing of value to preserve.

## 2. What We Already Tried (and why it stopped)

| Attempt                                             | Result                                                                                                                   |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>CM Resolve</b> (KB 439836)                       | <b>Failed</b> — re-runs a clear-ownership call that can't complete on a deleted object.                                  |
| <b>Remove NSX</b> from cluster (KB 319037 Option 2) | <b>Stalled at 20%</b> — can't remove a host-switch that points at the missing VDS (same wall). Hosts remained up/online. |

The clean, self-service UI paths are **exhausted** — both blocked on the same deleted-VDS reference.

## 3. Recovery Options (Honest Assessment)

| #        | Option                                                                  | Prescribed?                 | Fixes it?              | Verdict                                                                                                      |
|----------|-------------------------------------------------------------------------|-----------------------------|------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>A</b> | <b>Broadcom Support</b> — Corfu DB cleanup of the stale ownership       | ✓ Prescribed route          | ✓ In place             | <b>The correct fix</b> — but <b>no support contract</b>                                                      |
| <b>B</b> | <b>Restore NSX from backup</b>                                          | ✓ (for Manager loss)        | ✗                      | Re-imports today's broken state; <b>does not fix</b>                                                         |
| <b>C</b> | <b>Redeploy a failed Manager node</b>                                   | ✓ (for node failure)        | ✗                      | <b>N/A</b> — no node failed; Manager is healthy                                                              |
| <b>D</b> | <b>Manual clean NSX rebuild</b> (tear down, deploy fresh, re-integrate) | ✗ <b>Not VCF-prescribed</b> | ✓ <i>if</i> complete d | <b>Improvised / out-of-VCF</b> ; desyncs SDDC Manager; high effort; <b>not a guess-free runbook</b> (see §5) |
| <b>E</b> | <b>Accept overlay-down; run on VLAN</b>                                 | n/a                         | — (avoids)             | <b>Zero risk</b> ; overlay isn't harming anything                                                            |

## 4. Recommendation

1. **Now: Option E** — run workloads (incl. the Aria workshop) on the **VLAN-backed network** `pg-vm-mgmt`. Nothing is degrading; this is free and reversible.
2. **To actually fix the overlay: Option A (Broadcom Support)** is the only *prescribed* in-place fix. If a support contract becomes available, that's the route.
3. **Only if overlay is genuinely required and support is unavailable: Option D (manual rebuild)** as a **deliberate, planned lab exercise** — executed step-by-step from the verified Broadcom NSX 9.0 docs, accepting the VCF-desync risk. **Do not improvise it.**

## 5. Option D — Manual Rebuild: Phases Only (NOT a runbook)

**This is a phase outline, not click-by-click.** Each phase must be executed from the **verified Broadcom NSX 9.0 Installation/Administration Guide** for that exact step, confirmed against the live UI/API before running. It is **outside VCF orchestration** — SDDC Manager will need separate reconciliation afterward (the same inventory issue that is already open). Treat as a major change with the NSX backup retained and a maintenance window.

1. **Tear down the stuck install.** The standard "Remove NSX" stalls here. Clearing it likely needs **force-removal of the transport nodes** and/or **Support/Corfu** — i.e., this very first phase may itself require Option A. *Do not force-delete blind.*
2. **Decide Manager disposition.** The Manager is healthy but holds the stale records. A truly clean result means a **fresh NSX deployment** (new appliance, **no restore** from the broken backup) — which forfeits the existing (empty) config and **detaches from VCF/SDDC management.**
3. **Deploy fresh NSX** (OVA, per the NSX 9.0 install guide) and form the cluster.
4. **Register the current vCenter** as Compute Manager (it will discover `vcenter-c101-vds01` / the current VDS).
5. **Recreate fabric objects** — uplink profile, TEP IP pool, transport zones (overlay + VLAN), and a Transport Node Profile bound to the **current VDS.**
6. **Prepare the cluster** (host transport nodes) on the current VDS — a *fresh* prepare, so no single-transaction relocation, no 9560.
7. **Verify** — transport nodes `success`, **tunnels UP (> 0)**, a test overlay segment passes traffic.
8. **Reconcile with VCF/SDDC Manager** — re-establish NSX in SDDC inventory (this is the open `NO_CLUSTERS` item; likely **Support-assisted**).

### 5.1 Verified host-cleanup commands — the Phase-1 unblock

**Verified 2026-06-09 (KB 322412 / 324239 / 375111).** Phase 1 is **no longer support-only** — the hosts can be cleaned **host-side** via the NSX CLI, one host at a time (host in **maintenance mode**; vMotion evacuates it — vMotion is healthy; **vLCM/image clusters reboot** after to drop VIBs):

```
del nsx          # on the ESXi host (nsxcli) – removes NSX host-switches + VIBs; watch /var/log/nsxcli.log
```

**If `del nsx` fails with "Unable to clear DVS property ... Busy":**

```
esxcli network ip interface remove --interface-name=vmk10  # remove the VTEP vmkernel
net-dvs -l          # find stale VTEP/vdrPort/spfPort UUIDs
net-dvs -D -p <port_uuid> <switch_name>                  # remove each stale port
del nsx             # retry
/usr/lib64/vmware/nsx-opsagent/bin/nsxcfg-vswitch -s      # force CVDS->DVS hot-swap, then:
del nsx
```

(Last resort: open `https://<esxi-host>/mob` and delete the VDS/NVDS object still carrying NSX properties.)

Once all four hosts are clean, choose a Manager disposition: - **Path A (reuse existing Manager — try first):** force-remove the now-orphaned transport nodes ( `DELETE /api/v1/transport-nodes/<id>?force=true` ), confirm the CM 903 25 error clears, then **re-prepare** on the current VDS (Configure NSX → TNP). Fresh prepare = no relocation, no 9560.

- **Path B (fresh Manager — if A stays wedged):** deploy a new NSX 9.0 Manager per the **NSX 9.0 Installation Guide**, register the current vCenter, recreate uplink profile + TEP IP pool + transport zones + TNP on the current VDS, then prepare the cleaned hosts.

**Phase 8 (re-syncing NSX into SDDC Manager inventory) is the only piece that still likely needs Broadcom Support.** A lab can accept a **functional-but-unmanaged** NSX (overlay works; VCF just doesn't track it).

## 6. Risks

| Risk                       | Note                                                                       |
|----------------------------|----------------------------------------------------------------------------|
| Teardown stuck (Phase 1)   | Same deleted-VDS wall; may require Support/force-removal                   |
| VCF desync deepened        | Manual NSX outside SDDC orchestration worsens the existing inventory break |
| Backup is not a clean undo | NSX restore = redeploy Manager + restore → returns to <i>broken</i> state  |
| Fragile nested lab         | Host-level NSX operations on a single vSAN cluster carry real risk         |

## 7. Decision

**Decision:** Run workloads on **VLAN ( pg-vm-mgmt )** now (Option E). Pursue the overlay fix via **Broadcom Support (Option A)** when available; that is the only prescribed in-place remedy. Undertake the **manual rebuild (Option D)** only as a deliberate, doc-driven lab exercise if overlay is required and support is not — accepting that Phases 1 and 8 may still need support and that it desyncs VCF. **Justification:** The clean self-service fixes are exhausted on the deleted-VDS reference; no VCF-prescribed from-scratch rebuild exists; the overlay outage is non-harmful (mgmt/VLAN healthy, config empty). **Implication:** Overlay stays down until Option A or D is executed; workloads use VLAN meanwhile; this remains entangled with the open SDDC-inventory and vCenter- **ACTIVATING** items.

## 8. Document Information

| Field          | Value                                               |
|----------------|-----------------------------------------------------|
| Title          | NSX Recovery Options & Clean-Rebuild Decision Plan  |
| Document ID    | VC-PLAN-NSX-RECOVERY-20260609                       |
| Version        | 1.1                                                 |
| Issued         | 2026-06-09                                          |
| Author         | Virtual Control LLC                                 |
| Classification | Internal                                            |
| Platform       | VMware Cloud Foundation 9.0.1 (nested) — NSX on VDS |

| Field             | Value                                                                                                                                                                                    |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Related Documents | PLAN-NSX-Transport-Node-VDS-Reconciliation-2026-06-09; VCF9-SDDC-Manager-Inventory-State-Known-Issues-2026-06-08; VCF9-Operations-Campaign-2026-06-06-to-2026-06-19                      |
| Distribution      | Lab notebook                                                                                                                                                                             |
| Key references    | KB 439836 (CM Resolve); KB 319037 (error 9560 uninstall/re-prepare); KB 425441 (post-uninstall stuck state); VCF 9.0 NSX backup/restore & redeploy-node docs; NSX 9.0 Installation Guide |

## Revision History

| Version | Date       | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | 2026-06-09 | Initial recovery decision plan. Established that <b>no VCF-prescribed from-scratch NSX rebuild exists</b> for this stale-VDS scenario (recovery docs are restore/redeploy-node only); documented the verified diagnosis, the exhausted self-service attempts (Resolve failed, Remove NSX stalled), the five recovery options with honest verdicts (Support = the only prescribed in-place fix), a phase-only outline of the eyes-open manual rebuild (Phases 1 & 8 likely still need Support), risks, and the decision to run on VLAN now. |
| 1.1     | 2026-06-09 | Added §5.1 <b>verified host-cleanup commands</b> ( <code>del nsx</code> + the "DVS property Busy" workaround — <code>esx cli network ip interface remove , net-dvs -l/-D , nsxcfg-vswitch -s</code> ; KB 322412/324239/375111). This <b>unblocks Phase 1 as self-service</b> (host-side cleanup, no Support needed) and reduces the residual Support dependency to <b>Phase 8 only</b> (re-syncing NSX into SDDC inventory). Confirmed a Manager reboot does <b>not</b> clean the hosts and would re-stall the 20% removal.                |

**Document End**